QF620 Stochastic Modelling in Finance — Session 3 Reference Cheatsheet

Brownian Motion and Martingale: random walks, the scaled random walk limit, the definition and properties of Brownian motion, Markov property, geometric Brownian motion, Brownian martingales

Built from Session 3 slides (Tee Chyng Wen, SMU) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Session 3: Brownian Motion and Martingale" (Tee Chyng Wen, SMU, 25 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded **where** box that names each symbol and component and says what it represents in that formula.

Exam context: the exam format is not stated on the Session 1–4 slides; check the course outline. From

this session you need: the four defining properties of Brownian motion and how to test a candidate process against them; mean, variance and covariance of W_t; probability calculations using independent increments; the Markov property; expectations of $e^{\sigma W_t}$ via the MGF; and proving or disproving the martingale property with $W_t = (W_t - W_s) + W_s$.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
S_n	position of a symmetric random walk after n steps	random	[S2]
X_i	i.i.d. ± 1 steps, probability $\frac{1}{2}$ each	random	[S9]
$W_n(t)$	scaled random walk with n steps on $[0, t]$	random	[S9]
W_t	(standard) Brownian motion / Wiener process at time t	random	[S11]
$s,\ t,\ T$	times, usually $0 \le s \le t \le T$	time	
$W_t - W_s$	increment of W over $[s, t]$	random	$\sim N(0, t-s)$ [S11]
$N(\mu, \sigma^2)$	normal distribution	distribution	
$\mathrm{Cov}(\cdot, \cdot)$	covariance	number	[S14]
$\mathbb{E}_t[\cdot]$	expectation conditional on information up to time t	random	[S16]
M_t	martingale	random	[S21]
$X_t = e^{W_t}$	geometric Brownian motion (slide's simplest form)	random	[S17]
σ	real constant (volatility)	number	[S24]

Notation trap: in Session 2 S_n was a stock price and X_i a coin toss; here S_n is the random-walk position on [S2] and X_t is a generic process. $\mathbb{E}_s[\cdot]$ means $\mathbb{E}[\cdot \mid \mathcal{F}_s]$, "given the path up to time s".

0. Topic map

```
PREREQUISITES
S1  normal distribution, MGF E[e^{θX}] = e^{μθ+½σ²θ²}, lognormal mean
S2  arithmetic tree ±σ√(T/n) + CLT ⇒ normal; conditional expectation rules; martingales on a tree
  |
  ▼
SESSION 3: BROWNIAN MOTION AND MARTINGALE
├─ BROWNIAN  [S2–S10]
|   ├─ Symmetric random walk: 2^n paths, range [−n, n]                    [S2]
|   ├─ History: Brown (pollen), Einstein, Bachelier 1900 (stock prices)   [S3–S8]
|   ├─ Scaled walk W_n: steps ±√(t/n) every t/n                           [S9]
|   └─ CLT ⇒ W_n(t) → N(0,t); independent increments ⇒ Brownian motion     [S10]
|
├─ DEFINITION OF W_t  [S11–S15]
|   ├─ W_0 = 0; continuous; stationary independent increments; W_t−W_s ~ N(0,t−s)  [S11]
|   ├─ √t Z: right marginals, wrong increments ⇒ not BM                   [S12–S13]
|   ├─ E[W_t]=0, E[W_t²]=t, Cov(W_s,W_t)=min(s,t); jointly normal         [S14]
|   └─ Examples: P(W₂<0 | W₁>0), P(W₁W₂>0)                                [S15]
|
├─ PROPERTIES  [S16–S20]
|   ├─ Markov: E_s[W_t] = W_s                                             [S16]
|   ├─ GBM X_t = e^{W_t}; E[X_T], E[X_T²]                                 [S17]
|   ├─ Nowhere differentiable, self-similar, fractal, recurrent           [S18–S19]
|   └─ Reflection −W_t is a Brownian motion                                [S20]
|
└─ MARTINGALES  [S21–S25]
    ├─ M_t = E_t[M_T]; W_t is a martingale                                [S21]
    ├─ W_t² − t is a martingale; W_t³ is not                              [S22–S23]
    ├─ E[e^{σW_t}] = e^{½σ²t}                                             [S24]
    └─ exp(σW_t − ½σ²t) is an exponential martingale                       [S25]
  |
  ▼
LATER: S4 SDEs dS = μ dt + σ dW, stochastic integrals, Itô's formula (box calculus dW² = dt);
       then Black–Scholes and Girsanov
```

Dependency logic:

- Session 2's arithmetic tree is exactly the scaled random walk; letting the step size go to zero gives Brownian motion, the continuous-time building block.
- The definition is about *increments*, not just marginals: that is what makes W Markov and lets you compute joint probabilities and conditional expectations by splitting $W_t = (W_t - W_s) + W_s$.
- The MGF from Session 1 gives $\mathbb{E}[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}$, which is why $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is a martingale: the continuous-time version of the $-\frac{1}{2}\sigma^2$ in the lognormal tree.
- Non-differentiability is why Session 4 needs stochastic integrals and Itô's formula instead of ordinary calculus.

1 Concept entries

1.1 Symmetric random walk [S2]

A 1-dimensional symmetric random walk starts at 0 and moves $+1$ or -1 with probability 0.5 each step. For 4 steps:

- Number of paths $= 2^4 = 16$; maximum 4, minimum -4; reachable values $-4, -2, 0, 2, 4$.

$$S_4 = 2B - 4, \qquad B \sim \text{Bin}(4,\, 0.5), \qquad \mathbb{P}(S_4 = 2k - 4) = \binom{4}{k}\left(\frac{1}{2}\right)^4.$$

where

S_4	position after 4 steps.
B	number of up steps among the 4 (binomial with 4 trials, success probability 0.5).
$2B - 4$	B ups and $4 - B$ downs give $B - (4 - B) = 2B - 4$.
$\binom{4}{k}(\frac{1}{2})^4$	probability of exactly k ups: $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$ for $S_4 = -4, -2, 0, 2, 4$.

△**!** [S2] writes $S_4 \sim \text{Bin}(4, 0.5)$. The *number of up-steps* is $\text{Bin}(4, 0.5)$; the position is the affine transform $2B - 4$ (mean 0, variance 4).

1.2 History: from pollen to stock prices [S3–S8]

- **Robert Brown** (1827) observed the erratic motion of pollen particles in water and concluded it came from the particles themselves, not currents or evaporation [S3].
- **Einstein** showed the motion follows a clear statistical law [S4].
- **Louis Bachelier**, *Théorie de la spéculation* (1900), modelled stock price fluctuations with Brownian motion, analysing forwards and options on the Paris exchange [S5–S8].
- Brownian motion remains the building block of the reference continuous-time market model; we approach it as an **infinitesimal random walk** with ever smaller, more frequent steps [S8].

1.3 From random walk to Brownian motion [S9–S10]

Scaled random walk $W_n(t)$ [S9]: (1) $W_n(0) = 0$; (2) time spacing t/n; (3) up and down jumps of size $\sqrt{t/n}$; (4) measure $\mathbb{P}$ with up/down probability $\frac{1}{2}$ everywhere.

$$W_n\Big(\frac{i}{n}\,t\Big) = W_n\Big(\frac{i-1}{n}\,t\Big) + \sqrt{\frac{t}{n}}\,X_i, \qquad 1 \le i \le n.$$

where

n	number of steps used to reach time t.
$\frac{i}{n}t$	time after i steps of length t/n.
X_i	independent coin toss: $+1$ or -1 with probability $\frac{1}{2}$; mean 0, variance 1.
$\sqrt{t/n}$	jump size; squared it equals the step length t/n, so variance accumulates at rate 1 per unit time.

Adding the n steps:

$$W_n(t) = \sqrt{t}\left(\frac{\sum_{i=1}^{n} X_i}{\sqrt{n}}\right) \xrightarrow[n\to\infty]{} \sqrt{t}\,Z \sim N(0,t).$$

where

$\sum_{i=1}^{n} X_i$	total number of ups minus downs.
$\frac{1}{\sqrt{n}}\sum X_i$	standardised sum (mean 0, variance 1) $\to N(0,1)$ by the central limit theorem [S10].
$\sqrt{t}$	rescales to variance t; the $\sqrt{t/n}$ scaling stops W_n from blowing up as n grows.

Why the limit is Brownian motion [S10]:

- All marginals and conditional marginals tend to normal distributions.
- Future moves away from any position are independent of where that position is and of the whole history up to then.
- A future displacement $W_n(s+t) - W_n(s)$ is (scaled) binomial with mean 0 and variance t; by the CLT it tends to $N(0,t)$.
- Hence **Brownian motion is the limit of a scaled random walk** as $n \to \infty$.

1.4 Mathematical definition of Brownian motion [S11]

Formalised by **Norbert Wiener** (1923). A process W_t is a Brownian motion (Wiener process) if and only if:

(1) $W_0 = 0$; (2) $t \mapsto W_t$ continuous (no jumps);
(3) stationary, independent increments; (4) $W_t - W_s \sim N(0,\, t-s),\ 0 \le s \le t$.

where

$W_0 = 0$	starts at the origin.
continuous	paths have no jumps.
independent increments	$W_{t_2} - W_{t_1}$, $W_{t_3} - W_{t_2}, \ldots$ over non-overlapping intervals are independent; in particular $W_t - W_s$ is independent of the whole path up to s.
stationary	the distribution of $W_t - W_s$ depends only on the length $t-s$, not on s.
$N(0, t-s)$	each increment is normal with mean 0 and variance equal to the elapsed time.

△! [S11] spells the name "Weiner"; it is Wiener.

1.5 Example: $\sqrt{t}\,Z$ is not a Brownian motion [S12–S13]

$Z \sim N(0,1)$, $X_t = \sqrt{t}\,Z$ is continuous and $X_t \sim N(0,t)$. Its increment uses the *same* Z:

$$X_{s+t} - X_s = \sqrt{s+t}\,Z - \sqrt{s}\,Z \sim \left(\sqrt{s+t} - \sqrt{s}\right)N(0,1) \sim N\Big(0,\ 2s + t - 2\sqrt{s(s+t)}\Big).$$

where

Z	one single standard normal draw, shared by all times.
$\sqrt{s+t} - \sqrt{s}$	the increment is a constant times Z.
$2s + t - 2\sqrt{s(s+t)}$	$(\sqrt{s+t} - \sqrt{s})^2$ expanded: the increment variance.

- Brownian motion needs increment variance t; here it is $2s + t - 2\sqrt{s(s+t)} < t$ for $s > 0$.
- The increment is a multiple of Z, as is X_s, so it is *not independent* of X_s (perfectly correlated).
- So X_t is **not** a Brownian motion: having $N(0,t)$ marginals is far weaker than having independent $N(0, t-s)$ increments [S13]. Its paths are deterministic curves $\sqrt{t}$ scaled by one draw, fanning out smoothly.

1.6 Statistical properties and joint normality [S14]

$$\mathbb{E}[W_t] = 0, \quad W_t \sim N(0,t), \qquad \mathbb{E}[W_t^2] = t,$$
$$\mathbb{E}[W_t - W_s] = 0, \qquad V[W_t - W_s] = \mathbb{E}[(W_t - W_s)^2] = t - s, \qquad \mathrm{Cov}(W_s, W_t) = s, \quad s < t.$$

where

$W_t \sim N(0,t)$	from property (4) with $s = 0$ and $W_0 = 0$.
$\mathbb{E}[W_t^2] = t$	variance t and mean 0.
$V[W_t - W_s]$	increment variance equals the elapsed time $t - s$.
$\mathrm{Cov}(W_s, W_t) = s$	*[Beyond slides]*: $\mathbb{E}[W_s W_t] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] = 0 + s$ by independent increments. In general $\mathrm{Cov}(W_s, W_t) = \min(s,t)$.

For $t_1 < t_2 < \cdots < t_n$, $(W_{t_1}, \ldots, W_{t_n})$ is **jointly normal**, mean 0, covariance matrix

$$\Sigma = \begin{bmatrix} t_1 & t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & t_2 & \cdots & t_2 \\ t_1 & t_2 & t_3 & \cdots & t_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & t_3 & \cdots & t_n \end{bmatrix}, \qquad \Sigma_{ij} = \min(t_i, t_j).$$

where

Σ	covariance matrix of the vector of Brownian values.
Σ_{ij}	covariance of W_{t_i} and W_{t_j}: the smaller of the two times.
diagonal	$\Sigma_{ii} = t_i$, the variances.

[Beyond slides] Correlation $\rho(W_s, W_t) = \frac{s}{\sqrt{st}} = \sqrt{s/t}$.

1.7 Probability examples [S15]

Write $W_2 = W_1 + Y$ with $Y = W_2 - W_1 \sim N(0,1)$ independent of $W_1 \sim N(0,1)$ (worked here, *[Beyond slides]*).
(a) Conditional on $W_1 > 0$, probability that $W_2 < 0$:

$$\mathbb{P}(W_2 < 0 \mid W_1 > 0) = \frac{\mathbb{P}(W_1 > 0,\ W_1 + Y < 0)}{\mathbb{P}(W_1 > 0)} = \frac{1/8}{1/2} = \frac{1}{4}.$$

where

$W_1,\ Y$	two independent standard normals: their joint density is rotationally symmetric.
$\{W_1 > 0,\ Y < -W_1\}$	a wedge of angle 45° in the (W_1, Y) plane, so probability $45/360 = \frac{1}{8}$.
$\mathbb{P}(W_1 > 0) = \frac{1}{2}$	symmetry of the normal.
$\frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$	definition of conditional probability.

(b) Probability that W_1 and W_2 have the same sign:

$$\mathbb{P}(W_1 W_2 > 0) = 1 - \mathbb{P}(W_1 > 0, W_2 < 0) - \mathbb{P}(W_1 < 0, W_2 > 0) = 1 - \tfrac{1}{8} - \tfrac{1}{8} = \tfrac{3}{4}.$$

where

$W_1 W_2 > 0$	both positive or both negative.
$\frac{1}{8}$	each sign-change event is a 45° wedge (case (a) and its mirror image by reflection symmetry).

[Beyond slides] General: $\mathbb{P}(W_s W_t > 0) = \frac{1}{2} + \frac{1}{\pi} \arcsin \sqrt{s/t}$; for $s = 1, t = 2$: $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ ✓.

1.8 Markov property [S16]

A process is **Markovian** if the conditional distribution of its future depends only on its present value:

$$\mathbb{E}_t[X_T] = \mathbb{E}[X_T \mid X_t, X_{t-1}, X_{t-2}, \ldots, X_0] = \mathbb{E}[X_T \mid X_t].$$

where

X_T	value at a future time $T > t$.
$X_t, \ldots, X_0$	the whole history up to time t.
$\mathbb{E}[X_T \mid X_t]$	conditioning on the present value alone gives the same answer: all relevant information is in the present state.

Brownian motion is Markov [S16] $(0 < s < t)$:

$$\mathbb{E}_s[W_t] = \mathbb{E}_s[W_t - W_s + W_s] = \mathbb{E}_s[W_t - W_s] + \mathbb{E}_s[W_s] = 0 + W_s = W_s.$$

where

$W_t - W_s + W_s$	add and subtract W_s to split into a future increment and a known value.
$\mathbb{E}_s[W_t - W_s] = 0$	the increment is independent of the past and has mean 0 (*independence*).
$\mathbb{E}_s[W_s] = W_s$	W_s is known at time s (*taking out what is known*).

To predict W_t given information up to s you only need W_s. All independent-increment processes are Markov.
△**!** [S16] states the Markov property with conditional *expectations*; the property is about the whole conditional *distribution* (it must hold for $\mathbb{E}[g(X_T) \mid \cdot]$ for every g). The expectation version is a consequence.

1.9 Geometric Brownian motion [S17]

$$X_t = e^{W_t}$$

where

W_t	Brownian motion.
X_t	geometric Brownian motion (GBM): its logarithm is a Brownian motion; always positive. Used for asset prices in the Black–Scholes model. *[Beyond slides]* The full model is $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$.

Slide example [S17] (worked here, *[Beyond slides]*; MGF with $W_T \sim N(0, T)$):

$$\mathbb{E}[X_T] = \mathbb{E}\big[e^{W_T}\big] = e^{\frac{1}{2}T}, \qquad \mathbb{E}[X_T^2] = \mathbb{E}\big[e^{2W_T}\big] = e^{\frac{1}{2}\cdot 4T} = e^{2T}.$$

where

$\mathbb{E}[e^{\theta W_T}] = e^{\frac{1}{2}\theta^2 T}$	normal MGF with $\mu = 0$, $\sigma^2 = T$.
$\theta = 1$	gives the mean.
$\theta = 2$	since $X_T^2 = e^{2W_T}$.

[Beyond slides] $V[X_T] = e^{2T} - e^T$. Note $\mathbb{E}[X_T] = e^{T/2}$ grows although $\mathbb{E}[W_T] = 0$: convexity.

1.10 Path properties [S18–S20]

- **Nowhere differentiable** [S18–S19]: smooth functions look like straight lines when you zoom in; Brownian paths stay just as rough at every scale. W_t is continuous everywhere but differentiable nowhere. *[Beyond slides]* Heuristic: $\frac{W_{t+h} - W_t}{h} \sim N(0, 1/h)$, whose variance explodes as $h \to 0$.
- **Self-similar / fractal** [S18–S19]: looks the same at every scale. *[Beyond slides]* Scaling: $\frac{1}{\sqrt{c}} W_{ct}$ is again a Brownian motion for any $c > 0$.
- **Recurrence** [S19]: it hits every real value eventually (with probability 1), returns to zero however far it wanders, and once it hits a value it hits it again infinitely often.
- **Reflection** [S20]: $W_t \mapsto -W_t$ is again a Brownian motion, because each increment's distribution $N(0, t-s)$ is symmetric about 0.

1.11 Brownian martingales [S21–S23]

Definition [S21]:

$$M_t = \mathbb{E}_t[M_T] \qquad \forall\, T \geq t.$$

where

M_t	a stochastic process (value at time t).
$\mathbb{E}_t[M_T]$	best forecast of a future value given information up to t.
$=$	the forecast of any future value is today's value: no expected drift. *[Beyond slides]* Also need $\mathbb{E}\lvert M_t\rvert < \infty$.

W_t is a martingale [S21] ($0 \leq s \leq t$):

$$\mathbb{E}_s[W_t] = \mathbb{E}_s[W_t - W_s + W_s] = \mathbb{E}_s[W_t - W_s] + \mathbb{E}_s[W_s] = W_s.$$

where

$\mathbb{E}_s[W_t - W_s] = 0$	increment independent of $\mathcal{F}_s$, mean 0.
$\mathbb{E}_s[W_s] = W_s$	known at s.

$W_t^2 - t$ is a martingale [S22] (steps filled in, *[Beyond slides]*):

$$\mathbb{E}_s[W_t^2 - t] = \mathbb{E}_s\big[(W_t - W_s + W_s)^2\big] - t = \underbrace{\mathbb{E}_s[(W_t - W_s)^2]}_{t-s} + \underbrace{2W_s \mathbb{E}_s[W_t - W_s]}_{0} + W_s^2 - t = W_s^2 - s.$$

where

$(W_t - W_s)^2$	squared increment: independent of the past, mean $t - s$.
$2W_s(W_t - W_s)$	cross term: W_s comes out (known), increment has mean 0.
W_s^2	known at s.
$-t$	deterministic; $t - s - t = -s$.
$W_s^2 - s$	the same process at time s: martingale ✓. Without the $-t$, W_t^2 drifts up at rate 1.

W_t^3 is not a martingale [S23]:

$$\mathbb{E}_s[W_t^3] = \mathbb{E}_s\big[(W_t - W_s)^3 + 3(W_t - W_s)^2 W_s + 3(W_t - W_s)W_s^2 + W_s^3\big] = 3(t-s)W_s + W_s^3.$$

where

$\mathbb{E}_s[(W_t - W_s)^3] = 0$	odd moment of a mean-zero normal.
$3(W_t - W_s)^2 W_s$	gives $3(t-s)W_s$.
$3(W_t - W_s)W_s^2$	has conditional mean 0.
$3(t-s)W_s$	the extra term: $\mathbb{E}_s[W_t^3] \neq W_s^3$, so not a martingale. *[Beyond slides]* $W_t^3 - 3tW_t$ *is* a martingale.

△**!** [S23] concludes that W_t^3 "doesn't satisfy the definition of a standard Brownian motion"; it should say it does not satisfy the definition of a *martingale*.

1.12 Exponential martingale [S24–S25]

Recall the MGFs [S24]:

$$X \sim N(\mu, \sigma^2):\ \mathbb{E}\big[e^{\theta X}\big] = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}, \qquad Z \sim N(0,1):\ \mathbb{E}\big[e^{\theta Z}\big] = e^{\frac{1}{2}\theta^2}.$$

where

θ	real number at which the MGF is evaluated.
$\mu,\ \sigma^2$	mean and variance of X.

Slide example [S24] (worked here, *[Beyond slides]*): $W_t \sim N(0,t)$, $\sigma \in \mathbb{R}$:

$$\mathbb{E}\big[e^{\sigma W_t}\big] = e^{\frac{1}{2}\sigma^2 t}.$$

where

σ	real constant playing the role of θ.
W_t	normal with mean 0, variance t; so $\mu\theta = 0$ and $\frac{1}{2}\sigma^2\theta^2 \to \frac{1}{2}t\sigma^2$.

Exponential martingale [S25]: $\exp\left(\sigma W_t - \frac{1}{2}\sigma^2 t\right)$ is a martingale.

$$\mathbb{E}_s\big[e^{\sigma W_t - \frac{1}{2}\sigma^2 t}\big] = \mathbb{E}_s\big[e^{\sigma(W_t - W_s)} \cdot e^{\sigma W_s - \frac{1}{2}\sigma^2 t}\big] = e^{\sigma W_s - \frac{1}{2}\sigma^2 t} \cdot \mathbb{E}_s\big[e^{\sigma(W_t - W_s)}\big] = e^{\sigma W_s - \frac{1}{2}\sigma^2 t} \cdot e^{\frac{1}{2}\sigma^2(t-s)} = e^{\sigma W_s - \frac{1}{2}\sigma^2 s}.$$

where

$e^{\sigma(W_t - W_s)}$	the future part: depends only on the increment, independent of the past.
$e^{\sigma W_s - \frac{1}{2}\sigma^2 t}$	known at s, so it comes out of $\mathbb{E}_s$.
$e^{\frac{1}{2}\sigma^2(t-s)}$	MGF of the increment $W_t - W_s \sim N(0, t-s)$.
$-\frac{1}{2}\sigma^2 t$	compensator: exactly cancels the convexity growth $e^{\frac{1}{2}\sigma^2 t}$, leaving no drift.

[Beyond slides] Finance link: the discounted risk-neutral stock $e^{-rt}S_t = S_0 e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is this martingale; it is the continuous-time version of Session 2's "discounted prices are $\mathbb{Q}$-martingales".

1.13 Implementation *[Beyond slides]* (tested)

```
import numpy as np
rng = np.random.default_rng(0)
n = 2_000_000
W1 = rng.standard_normal(n)
W2 = W1 + rng.standard_normal(n)            # W2 - W1 ~ N(0,1), independent of W1
print(round(np.mean(W2[W1 > 0] < 0), 3))    # P(W2<0 | W1>0) = 0.25
print(round(np.mean(W1*W2 > 0), 3))         # P(W1 W2 > 0)   = 0.75
T = 1.0
WT = np.sqrt(T) * rng.standard_normal(n)
print(round(np.mean(np.exp(WT)), 3), round(np.exp(T/2), 3))      # 1.649 1.649
print(round(np.mean(np.exp(2*WT)), 2), round(np.exp(2*T), 2))    # 7.41 7.39 (MC noise)
```

2 Comparison tables

2.1 Brownian motion vs $\sqrt{t}\,Z$ [S12–S13]

	Brownian motion W_t	$X_t = \sqrt{t}\,Z$
Marginal at t	$N(0,t)$	$N(0,t)$
Increment $[s, s+t]$	$N(0,t)$	$N\big(0, (\sqrt{s+t}-\sqrt{s})^2\big)$
Increment vs past	independent	perfectly correlated
Paths	rough, nowhere differentiable	smooth curves
Brownian motion?	yes	no

2.2 Random walk vs Brownian motion

	Scaled random walk $W_n(t)$	Brownian motion W_t
Time	discrete, step t/n	continuous
Step	$\pm\sqrt{t/n}$, probability $\frac{1}{2}$	$\mathrm{d}W_t$, infinitesimal
Distribution at t	scaled binomial, mean 0, var t	$N(0,t)$ exactly
Increments	independent	independent, stationary, normal
Link	$W_n(t) \to W_t$ as $n \to \infty$ (CLT)	

2.3 Martingale or not?

Process	$\mathbb{E}_s[\cdot]$	Martingale?
W_t	W_s	yes [S21]
W_t^2	$W_s^2 + (t-s)$	no (drift +1)
$W_t^2 - t$	$W_s^2 - s$	yes [S22]
W_t^3	$W_s^3 + 3(t-s)W_s$	no [S23]
$W_t^3 - 3tW_t$	$W_s^3 - 3sW_s$	yes *[Beyond slides]*
$e^{\sigma W_t}$	$e^{\sigma W_s + \frac{1}{2}\sigma^2(t-s)}$	no
$e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$	$e^{\sigma W_s - \frac{1}{2}\sigma^2 s}$	yes [S25]

2.4 Path properties

Continuity	continuous everywhere [S11, S19]
Differentiability	nowhere differentiable [S18–S19]
Scale	self-similar, fractal [S19]
Recurrence	hits every level; returns to 0 infinitely often [S19]
Invariance	reflection $-W_t$ [S20]; scaling $c^{-1/2}W_{ct}$ *[Beyond slides]*

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Random walk and limit [S2, S9–S10]

- $W_n(\frac{i}{n}t) = W_n(\frac{i-1}{n}t) + \sqrt{t/n}\,X_i$; $W_n(t) = \sqrt{t}\,\frac{\sum X_i}{\sqrt{n}} \to N(0,t)$.
  where $X_i = \pm 1$ with probability $\frac{1}{2}$; n = steps; t/n = step length.

Definition and moments [S11–S14]

- $W_0 = 0$; continuous; independent stationary increments; $W_t - W_s \sim N(0, t-s)$.
  where $0 \le s \le t$.
- $\mathbb{E}[W_t] = 0$, $\mathbb{E}[W_t^2] = t$, $V[W_t - W_s] = t - s$, $\mathrm{Cov}(W_s, W_t) = \min(s,t)$, $\rho = \sqrt{s/t}$.
  where Cov = covariance; ρ = correlation ($s < t$).

Probabilities [S15]

- $\mathbb{P}(W_2 < 0 \mid W_1 > 0) = \frac{1}{4}$; $\mathbb{P}(W_1 W_2 > 0) = \frac{3}{4}$; generally $\mathbb{P}(W_s W_t > 0) = \frac{1}{2} + \frac{1}{\pi}\arcsin\sqrt{s/t}$ *[Beyond slides]*.
  where split $W_2 = W_1 + (W_2 - W_1)$ into independent normals.

Markov and martingales [S16, S21–S25]

- Markov: $\mathbb{E}_t[X_T] = \mathbb{E}[X_T \mid X_t]$; $\mathbb{E}_s[W_t] = W_s$.
  where $\mathbb{E}_s$ = expectation given information up to s.
- Martingale: $M_t = \mathbb{E}_t[M_T]$. Martingales: W_t, $W_t^2 - t$, $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$. Not: W_t^3 ($\mathbb{E}_s = W_s^3 + 3(t-s)W_s$).
  where σ = real constant.

Exponentials [S17, S24]

- $\mathbb{E}[e^{\theta X}] = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$; $\mathbb{E}[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}$; $\mathbb{E}[e^{W_T}] = e^{T/2}$, $\mathbb{E}[e^{2W_T}] = e^{2T}$.
  where $X \sim N(\mu, \sigma^2)$; $W_t \sim N(0,t)$; $X_t = e^{W_t}$ = GBM.

4 Glossary

- **Brownian motion / Wiener process:** continuous process with $W_0 = 0$ and independent $N(0, t-s)$ increments [S11].
- **Exponential martingale:** $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ [S25].
- **Fractal / self-similar:** looks statistically the same at every scale [S19].
- **Geometric Brownian motion (GBM):** exponential of a Brownian motion; Black–Scholes price model [S17].
- **Increment:** $W_t - W_s$ [S11].
- **Independent increments:** increments over disjoint intervals are independent [S11].
- **Jointly normal:** any vector of W_{t_i} is multivariate normal [S14].
- **Markov property:** the future depends on the past only through the present [S16].
- **Martingale:** $M_t = \mathbb{E}_t[M_T]$ [S21].
- **Nowhere differentiable:** no derivative at any time [S19].
- **Recurrence:** returns to every level infinitely often [S19].
- **Reflection invariance:** $-W_t$ is a Brownian motion [S20].
- **Scaled random walk:** steps $\pm\sqrt{t/n}$ every t/n [S9].
- **Stationary increments:** distribution depends only on the interval length [S11].
- **Symmetric random walk:** ± 1 steps with probability $\frac{1}{2}$ [S2].

5 Self-test

Questions

Q1 ([S2]). For a 4-step symmetric random walk, give the distribution, mean and variance of S_4.

Q2 ([S9–S10]). Why are the scaled walk's jumps $\pm\sqrt{t/n}$ and not $\pm t/n$?

Q3 ([S11]). State the four defining properties of Brownian motion.

Q4 ([S12]). Is $X_t = \sqrt{t}\,Z$ a Brownian motion? Is $\tilde{W}_t = -W_t$? Is $\frac{1}{\sqrt{c}}W_{ct}$?

Q5 ([S14]). Show $\mathrm{Cov}(W_s, W_t) = s$ for $s < t$ and write the covariance matrix of (W_1, W_2, W_3).

Q6 ([S15]). Find $\mathbb{P}(W_2 < 0 \mid W_1 > 0)$ and $\mathbb{P}(W_1 W_2 > 0)$.

Q7 ([S17]). For $X_t = e^{W_t}$ find $\mathbb{E}[X_T]$, $\mathbb{E}[X_T^2]$ and $V[X_T]$.

Q8 ([S21–S23]). Show W_t and $W_t^2 - t$ are martingales and W_t^3 is not. Find the compensated martingale built from W_t^3.

Q9 ([S24–S25]). Find $\mathbb{E}[e^{\sigma W_t}]$ and show $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is a martingale.

Q10 ([S16]). What is the Markov property and why does Brownian motion have it?

Solutions

S1. $S_4 = 2B - 4$, $B \sim \text{Bin}(4, \frac{1}{2})$: $\mathbb{P}(S_4 = -4, -2, 0, 2, 4) = \frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$. Mean 0, variance 4.

S2. The variance of the sum is $n \times (\text{jump})^2$; with jump $\sqrt{t/n}$ this is t. With t/n it would be $t^2/n \to 0$ (no randomness left).

S3. $W_0 = 0$; continuous paths; stationary, independent increments; $W_t - W_s \sim N(0, t - s)$.

S4. $\sqrt{t}Z$: no (wrong increment variance and increments not independent). $-W_t$: yes (reflection). $c^{-1/2}W_{ct}$: yes; increments $N(0, \frac{c(t-s)}{c}) = N(0, t - s)$, independent.

S5. $\mathbb{E}[W_s W_t] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] = 0 + s$. $\Sigma = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$.

S6. $W_2 = W_1 + Y$; the region $\{W_1 > 0, Y < -W_1\}$ is a 45° wedge: $\frac{1}{8}$. Conditional: $\frac{1/8}{1/2} = \frac{1}{4}$. Same sign: $1 - 2 \cdot \frac{1}{8} = \frac{3}{4}$.

S7. $e^{T/2}$, e^{2T}, $e^{2T} - e^T$.

S8. See §1.11. $\mathbb{E}_s[W_t^3] = W_s^3 + 3(t - s)W_s$; $\mathbb{E}_s[3tW_t] = 3tW_s$, so $W_t^3 - 3tW_t \to W_s^3 - 3sW_s$: martingale.

S9. $e^{\frac{1}{2}\sigma^2 t}$ (MGF with variance t). Split off the increment, pull out the known factor, apply the MGF to $W_t - W_s$: get $e^{\sigma W_s - \frac{1}{2}\sigma^2 s}$.

S10. The conditional distribution of the future depends only on the present value. For W: $W_t = W_s + (W_t - W_s)$ and the increment is independent of the whole past, so only W_s matters.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	4-step symmetric random walk	1.1
S3	Robert Brown quotation; sample paths	1.2
S4	C. P. Snow quotation: Einstein's statistical law	1.2
S5	Bachelier quotation and translation	1.2
S6	Bachelier: stock exchange operations	1.2
S7	Bachelier: *Théorie de la spéculation* figures	1.2
S8	Brownian motion as model of stock prices; infinitesimal random walk	1.2
S9	Scaled random walk $W_n(t)$	1.3
S10	CLT limit; independent increments; BM as limit	1.3
S11	Mathematical definition of Brownian motion	1.4
S12	Example: $\sqrt{t}Z$ is not BM	1.5
S13	Marginals vs increments; path comparison	1.5
S14	Moments, covariance, jointly normal	1.6
S15	Probability examples	1.7
S16	Markov property	1.8
S17	Geometric Brownian motion; example	1.9
S18	Differentiability (pictures)	1.10
S19	Fractal properties	1.10
S20	Reflection	1.10
S21	Martingale definition; W_t martingale	1.11
S22	$W_t^2 - t$ martingale	1.11
S23	W_t^3 not a martingale	1.11
S24	MGF recap; $\mathbb{E}[e^{\sigma W_t}]$	1.12

Slide	Content	Section
S25	Exponential martingale	1.12

Appendix B. Flag summary (△!)

1. [S2] $S_4 \sim \text{Bin}(4, 0.5)$: the number of ups is binomial; $S_4 = 2B - 4$.
2. [S11] "Weiner" should be Wiener.
3. [S16] The Markov property concerns conditional distributions; the slide's expectation form is a consequence.
4. [S21] The martingale definition also needs integrability $\mathbb{E}|M_t| < \infty$ and adaptedness (standard, omitted).
5. [S23] "doesn't satisfy the definition of a standard Brownian motion" should read "of a martingale".
6. [S15, S17, S24] Examples are posed without worked answers; the solutions here are my own.